

Q1) a) What is Model Building elaborate this phase of data analytics with the help of suitable example.

Ans:- **Model building** is a crucial phase in data analytics where machine learning models are trained using historical data to make predictions or decisions on new, unseen data. This phase involves selecting appropriate algorithms, training the models, tuning hyperparameters, evaluating performance, and deploying the models for real-world use.

Here's an elaboration of the model building phase with a suitable example:

****Example: Predicting Housing Prices****

Imagine you're working for a real estate company tasked with building a model to predict housing prices based on various features such as location, size, number of bedrooms, etc.

- 1. **Data Collection and Preprocessing**:** The first step is to gather a dataset containing historical housing data, including features like square footage, number of bedrooms, location, amenities, and selling prices. This data might come from various sources such as real estate databases, government records, or online listings. Preprocessing involves cleaning the data, handling missing values, encoding categorical variables, and splitting the data into training and testing sets.
- 2. **Feature Engineering**:** Feature engineering involves selecting, transforming, and creating new features from the raw data to improve the model's performance. For example, you might create a new feature by combining the number of bedrooms and bathrooms to represent the total number of rooms in the house.
- 3. **Model Selection**:** In this phase, you choose the appropriate machine learning algorithm(s) to train on the data. For predicting housing prices, regression algorithms such as linear regression, decision trees, random forests, or gradient boosting might be suitable choices.
- 4. **Training the Model**:** Once the algorithm is selected, the model is trained using the training data. During training, the model learns the patterns and relationships between the input features and the target variable (housing prices) from the historical data.
- 5. **Hyperparameter Tuning**:** Many machine learning algorithms have hyperparameters that need to be tuned to optimize the model's performance. Techniques like grid search or random search can be used to find the best combination of hyperparameters.
- 6. **Model Evaluation**:** After training and tuning the model, it's essential to evaluate its performance using the testing dataset. Common evaluation metrics for regression tasks include mean squared error (MSE), mean absolute error (MAE), and R-squared. The model's performance on the testing data helps assess its ability to generalize to new, unseen data.
- 7. **Deployment and Monitoring**:** Once the model has been trained and evaluated, it can be deployed into production for making predictions on new housing listings. It's crucial to monitor



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the model's performance over time and retrain it periodically with updated data to maintain its accuracy and reliability.

b) Explain any three sources of Big Data. Differentiate BI versus Data science.

Ans:-

****Three Sources of Big Data:****

1. **Transactional Data:** This type of data originates from various transactions that occur within organizations or between organizations and customers. It includes sales transactions, financial transactions, online purchases, and any other interactions that result in data being generated. Transactional data often contains structured information, such as customer IDs, timestamps, and transaction amounts.

2. **Social Media Data:** With the proliferation of social media platforms, vast amounts of data are generated every second in the form of posts, comments, likes, shares, and multimedia content. Social media data provides valuable insights into customer sentiments, preferences, trends, and interactions. Analyzing social media data can help businesses understand their target audience better, improve marketing strategies, and enhance customer engagement.

3. **Sensor Data:** Sensor data is generated by various sensors embedded in devices, machinery, vehicles, infrastructure, and IoT (Internet of Things) devices. These sensors collect real-time data on parameters such as temperature, pressure, humidity, location, motion, and environmental conditions. Sensor data is often unstructured or semi-structured and can be massive in volume. Analyzing sensor data enables organizations to monitor equipment health, detect anomalies, optimize processes, and make data-driven decisions in fields such as manufacturing, healthcare, transportation, and smart cities.

****Difference between Business Intelligence (BI) and Data Science:****

****1. Scope and Purpose:****

- ****Business Intelligence (BI)****: BI focuses on analyzing past and present data to provide descriptive insights into business operations. It typically involves querying, reporting, and visualization tools to generate reports, dashboards, and scorecards that help business users monitor key performance indicators (KPIs), track trends, and make data-driven decisions. BI is primarily used for monitoring business performance, identifying opportunities, and optimizing processes.

- ****Data Science****: Data science encompasses a broader range of techniques and methods for extracting insights and knowledge from data. It involves not only descriptive analytics (understanding what happened) but also predictive analytics (forecasting what will happen) and prescriptive analytics (suggesting what actions to take). Data science employs advanced

statistical analysis, machine learning, and data mining algorithms to uncover hidden patterns, make predictions, and generate actionable insights that drive innovation, optimize strategies, and create value.

****2. Data Types and Complexity:****

- ****BI****: BI primarily deals with structured data stored in relational databases, data warehouses, or OLAP (Online Analytical Processing) cubes. It focuses on aggregating and summarizing data from multiple sources to provide a consolidated view of business performance. BI tools are designed for handling structured data and generating predefined reports and visualizations quickly and efficiently.
- ****Data Science****: Data science works with both structured and unstructured data, including text, images, videos, and sensor data. Data scientists often deal with large, complex datasets (i.e., big data) that require specialized tools and techniques for data preprocessing, feature engineering, and model training. Data science projects involve exploratory data analysis, feature selection, model building, and evaluation, often requiring programming skills in languages like Python or R and knowledge of machine learning algorithms and statistical methods.

****3. Time Horizon and Decision-Making:****

- ****BI****: BI focuses on providing near real-time or historical insights into business operations, enabling operational reporting, performance monitoring, and trend analysis. BI reports and dashboards are often used by operational and middle-level managers to track KPIs, identify bottlenecks, and make short-term tactical decisions to improve efficiency and productivity.
- ****Data Science****: Data science aims to uncover actionable insights and predictive models that drive strategic decision-making and long-term planning. Data scientists develop models and algorithms that can forecast future trends, identify emerging patterns, and support strategic initiatives such as product development, market expansion, risk management, and customer segmentation. Data science solutions are often deployed as part of decision support systems or integrated into business processes to optimize outcomes and gain competitive advantages.

Q2) a) What are the three characteristic of Big Data and what are the main consideration in processing Big Data.

Ans:- The three main characteristics of Big Data, often referred to as the "3 Vs," are Volume, Variety, and Velocity:

1. **Volume:**

- ****Definition****: Refers to the vast amount of data generated from various sources, including transactions, social media interactions, sensor data, and more.

- **Significance**: Big Data involves datasets that are too large to be processed using traditional data processing techniques. It requires scalable infrastructure and storage solutions to handle and analyze these massive volumes of data effectively.

2. **Variety**:

- **Definition**: Refers to the diverse types and formats of data, including structured, semi-structured, and unstructured data.

- **Significance**: Big Data encompasses different types of data such as text, images, videos, sensor readings, social media posts, and more. Analyzing diverse data types requires flexible data processing techniques and tools capable of handling this variety effectively.

3. **Velocity**:

- **Definition**: Refers to the speed at which data is generated, collected, and processed in real-time or near real-time.

- **Significance**: Big Data often involves data streams generated continuously and at high speeds, such as social media updates, sensor readings, financial transactions, and website clickstreams. Processing and analyzing streaming data in real-time or near real-time enable organizations to make timely decisions, detect anomalies, and respond to events quickly.

Main considerations in processing Big Data include:

1. **Scalability**: Big Data processing systems need to be scalable to handle increasing volumes of data efficiently. This requires distributed computing frameworks and storage solutions that can scale horizontally by adding more resources (e.g., servers, storage nodes) as needed.

2. **Parallel Processing**: Processing large volumes of data in parallel is essential for achieving high performance and reducing processing times. Parallel processing techniques, such as MapReduce and parallel computing frameworks like Apache Hadoop and Apache Spark, enable data to be divided into smaller chunks and processed simultaneously across multiple nodes or processors.

3. **Data Integration and Interoperability**: Big Data often comes from disparate sources and in different formats. Effective data integration and interoperability are crucial for combining, cleansing, and transforming data from various sources into a unified format suitable for analysis. This involves using ETL (Extract, Transform, Load) processes, data integration tools, and standards to ensure data consistency and quality.

4. **Data Storage and Management**: Big Data requires robust storage and management solutions capable of storing, retrieving, and processing large volumes of data efficiently. This includes distributed file systems, NoSQL databases, and data warehouses optimized for handling Big Data workloads and supporting high-throughput data access.

5. ****Data Security and Privacy****: Big Data processing systems must address security and privacy concerns to protect sensitive data from unauthorized access, breaches, and misuse. This involves implementing access controls, encryption, authentication mechanisms, and compliance with data protection regulations (e.g., GDPR, HIPAA) to ensure data security and privacy.

By addressing these considerations, organizations can effectively process, analyze, and derive insights from Big Data to gain valuable business insights, optimize operations, and drive innovation.

b) Explain Descriptive, Diagnostic, Predictive analytics.

Ans:-

****Descriptive Analytics:****

Descriptive analytics focuses on summarizing and interpreting historical data to understand what has happened in the past. It involves analyzing data to describe and summarize key characteristics, trends, and patterns. Descriptive analytics answers questions like "What happened?" and "What is the current state of affairs?"

****Example****:

Imagine a retail store analyzing its sales data from the past year to understand monthly sales trends, peak sales hours, bestselling products, and customer demographics. By visualizing this data through charts, graphs, and dashboards, the store can gain insights into historical performance and make informed decisions about inventory management, marketing strategies, and customer engagement.

****Diagnostic Analytics:****

Diagnostic analytics delves deeper into historical data to identify the root causes of past events or outcomes. It involves examining patterns and relationships within the data to understand why certain events occurred. Diagnostic analytics answers questions like "Why did it happen?" and "What were the contributing factors?"

****Example****:

Continuing with the retail store example, suppose the store notices a sudden decline in sales during a particular month. Diagnostic analytics would involve analyzing additional data, such as marketing campaigns, competitor activities, economic factors, and customer feedback, to identify the reasons behind the sales decline. By pinpointing the contributing factors, the store can take corrective actions to address underlying issues and improve future performance.

****Predictive Analytics:****

Predictive analytics uses historical data and statistical algorithms to forecast future outcomes or trends. It involves building predictive models that can make predictions based on patterns observed in historical data. Predictive analytics answers questions like "What is likely to happen?" and "What are the potential outcomes?"

****Example****:

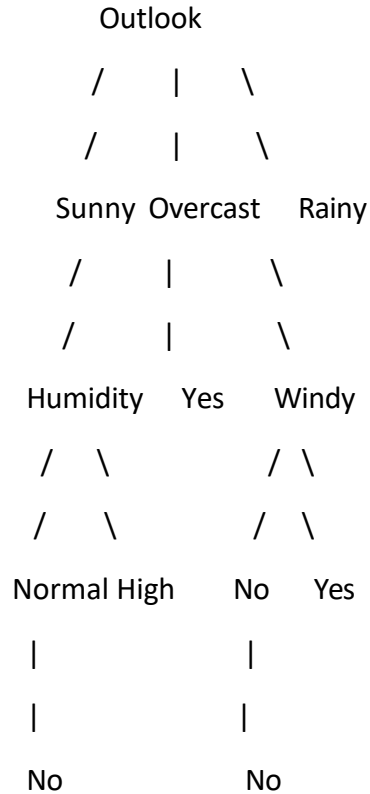
In the retail context, predictive analytics could involve building a sales forecasting model to predict future sales volumes based on historical sales data, seasonality, promotions, and other relevant factors. By leveraging machine learning algorithms such as linear regression, decision trees, or neural networks, the model can generate forecasts for future sales, helping the store anticipate demand, optimize inventory levels, and plan resources more effectively.

Q3) a) Explain why decision tree are used. Draw a sample decision tree and explain its parts.

Ans:- Decision trees are widely used in machine learning and data mining for classification and regression tasks. They are popular because of their simplicity, interpretability, and ability to handle both numerical and categorical data. Here's why decision trees are used:

- 1. **Interpretability****: Decision trees provide a clear and easy-to-understand representation of decision-making processes. Each node in the tree represents a decision based on a feature, and each branch represents an outcome or decision path. This makes decision trees particularly useful for explaining the logic behind a model's predictions to non-technical stakeholders.
- 2. **Versatility****: Decision trees can handle both classification and regression tasks, making them versatile for a wide range of predictive modeling problems. They can accommodate various types of input features, including numerical, categorical, and ordinal variables.
- 3. **Automatic Feature Selection****: Decision trees implicitly perform feature selection by identifying the most informative features at each decision point. This can help streamline the modeling process by focusing on the most relevant features and reducing the dimensionality of the problem.
- 4. **Non-parametric****: Decision trees are non-parametric models, meaning they make minimal assumptions about the underlying distribution of the data. This makes them robust to outliers and resistant to overfitting, especially when combined with techniques like pruning and ensemble methods.

Now, let's draw a sample decision tree and explain its parts:



****Parts of the Decision Tree:****

- 1. **Root Node (Outlook)**:** The topmost node of the tree represents the feature that best splits the dataset into distinct classes or groups. In this example, the Outlook is chosen as the root node because it provides the most significant information gain.
- 2. **Internal Nodes (Humidity, Windy)**:** Internal nodes represent decision points based on feature values. Each internal node tests a specific feature's value to determine the next node along the decision path.
- 3. **Leaf Nodes (Yes, No)**:** Leaf nodes represent the final decision or prediction for a particular class or outcome. Each leaf node corresponds to a specific class label or regression value.
- 4. **Branches**:** Branches emanating from each internal node represent the possible outcomes or decisions based on the feature's value being tested. Each branch leads to another node or leaf node, depending on the decision path.
- 5. **Decision Path**:** The path from the root node to a leaf node represents the sequence of decisions made by the decision tree to arrive at a final prediction or classification for a given instance.

b) How Apriori Algorithm works, explain with suitable example?

Ans:- The Apriori algorithm is a popular algorithm used for frequent itemset mining and association rule learning in data mining and machine learning. It is primarily employed in market basket analysis to identify frequent itemsets from transactional datasets and discover associations or relationships between items.

Here's how the Apriori algorithm works, explained with a suitable example:

****Example: Market Basket Analysis****

Suppose we have a transactional dataset containing records of customer purchases at a grocery store. Each transaction lists the items purchased by a customer in a single shopping trip.

Transaction 1: {bread, milk, eggs}

Transaction 2: {bread, butter}

Transaction 3: {milk, butter}

Transaction 4: {bread, milk, butter}

Transaction 5: {bread, milk, eggs, butter}

****Step 1: Find frequent itemsets of length 1 (Singletons):****

In this step, we identify individual items that occur frequently in the dataset. An itemset is considered frequent if it meets a specified minimum support threshold.

Support Threshold = 3 (minimum support count)

Frequent 1-itemsets:

{bread}, {milk}, {butter}, {eggs}

****Step 2: Generate candidate itemsets of length 2 (Pairs):****

Next, we generate candidate itemsets of length 2 by combining the frequent 1-itemsets found in the previous step. Only those candidate itemsets are considered which have all their subsets as frequent.

Candidate 2-itemsets:

{bread, milk}, {bread, butter}, {milk, butter}, {bread, eggs}

****Step 3: Calculate support for candidate itemsets of length 2:****

We scan the transactional dataset to count the occurrences of each candidate itemset. If the count of a candidate itemset meets the minimum support threshold, it is considered a frequent itemset.

Frequent 2-itemsets:

{bread, milk} (Support = 3), {bread, butter} (Support = 2), {milk, butter} (Support = 2)

****Step 4: Generate candidate itemsets of length 3 (Triples):****

Using the frequent 2-itemsets from the previous step, we generate candidate itemsets of length 3 by combining pairs of frequent itemsets. Again, we only consider candidate itemsets whose subsets are all frequent.

Candidate 3-itemsets:

{bread, milk, butter}

****Step 5: Calculate support for candidate itemsets of length 3:****

Similar to step 3, we count the occurrences of each candidate itemset in the dataset and identify the frequent itemsets.

Frequent 3-itemsets:

{bread, milk, butter} (Support = 2)

****Step 6: Generate association rules:****

Finally, we generate association rules from the frequent itemsets discovered in the previous steps. An association rule has the form $A \rightarrow B$, where A and B are itemsets, and the rule implies that if A is present in a transaction, then B is likely to be present as well.

Association rules:

{bread, milk} $\Rightarrow$ {butter}

{bread, butter} $\Rightarrow$ {milk}

{milk, butter} $\Rightarrow$ {bread}

These association rules provide insights into the purchasing behavior of customers. For example, based on the discovered association rule {bread, milk} $\Rightarrow$ {butter}, if a customer purchases bread and milk, there is a high likelihood that they will also purchase butter. Such insights can be leveraged for targeted marketing, product placement, and promotional strategies in the grocery store.

Q4) a) What is data preprocessing? Explain in details about handling missing data and transformation of data.

Ans:- Data preprocessing is a critical step in data analysis and machine learning pipelines that involves cleaning, transforming, and preparing raw data for analysis or model training. It aims to improve data quality, remove inconsistencies, and make the data suitable for further analysis or modeling

****Handling Missing Data:****

Missing data is a common issue in real-world datasets, and dealing with it appropriately is essential to prevent bias and ensure the reliability of analysis or modeling results. Several techniques can be used to handle missing data:

1. ****Dropping missing values****: One approach is to simply remove rows or columns with missing values from the dataset. While this can be effective, it may result in loss of valuable information, especially if the missing values are not randomly distributed.
2. ****Imputation****: Imputation involves replacing missing values with estimated values based on the available data. Common imputation techniques include:
 - Mean or median imputation: Replace missing values with the mean or median of the corresponding feature.
 - Mode imputation: Replace missing categorical values with the mode (most frequent value) of the feature.
 - Predictive imputation: Use predictive models to estimate missing values based on other features in the dataset.
3. ****Indicator variables****: Create binary indicator variables to flag missing values in the dataset. This approach allows the missingness pattern to be included as a feature in the analysis or modeling process.
4. ****Domain-specific knowledge****: In some cases, domain-specific knowledge or business rules can be used to infer or estimate missing values more accurately.

****Transformation of Data****

Data transformation involves modifying the original dataset to improve its distribution, scale, or relationship with the target variable. Common techniques for data transformation include:

1. ****Normalization****: Normalization rescales the features to a common scale, typically between 0 and 1 or -1 and 1. It helps to mitigate the effect of varying scales among features and ensures that all features contribute equally to the analysis or modeling process.
2. ****Standardization****: Standardization (or z-score normalization) transforms the features to have a mean of 0 and a standard deviation of 1. It centers the data around the mean and adjusts the scale to have unit variance, making it suitable for algorithms that assume normally distributed data.
3. ****Log Transformation****: Log transformation is used to stabilize the variance of features that exhibit heteroscedasticity (varying variance across different levels of the independent variable). It can also help to make skewed distributions more symmetric, making them suitable for certain statistical tests or modeling techniques.

4. **Encoding Categorical Variables:** Categorical variables are often encoded into numerical values before being used in machine learning algorithms. Common encoding techniques include one-hot encoding, label encoding, and target encoding, depending on the nature of the categorical variable and the requirements of the analysis or modeling task.

5. **Feature Engineering:** Feature engineering involves creating new features from existing ones to capture more complex relationships or patterns in the data. This may include polynomial features, interaction terms, or transformation of variables to capture nonlinear relationships.

Q5) a) What is text processing? Explain TF-IDF with example

Ans:- Text processing refers to the process of transforming and manipulating text data to extract useful information, analyze content, or prepare it for further analysis using computational methods. It involves several steps such as text cleaning, tokenization, normalization, and feature extraction.

TF-IDF (Term Frequency-Inverse Document Frequency) is a widely used technique in text processing and information retrieval to quantify the importance of a term in a document relative to a collection of documents. It aims to identify the most relevant terms in a document based on their frequency and rarity across the entire document collection.

Here's how TF-IDF works, explained with an example:

Consider a collection of documents (corpus) containing the following three documents:

Document 1: "The cat sat on the mat."

Document 2: "The dog chased the cat."

Document 3: "The mat was comfortable for the cat to sleep on."

Step 1: Tokenization and Preprocessing:

First, we tokenize each document into individual words and preprocess them by converting them to lowercase and removing punctuation.

Step 2: Calculate Term Frequency (TF):

Term Frequency (TF) measures how often a term occurs in a document relative to the total number of terms in that document. It is calculated using the formula:

$$TF(t, d) = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total number of terms in document } d}$$

For example, let's calculate TF for the term "cat" in Document 1:

$$TF("cat", \text{Document 1}) = \frac{1}{6} = 0.167$$

Similarly, we calculate TF for all terms in each document.

Step 3: Calculate Inverse Document Frequency (IDF):

Inverse Document Frequency (IDF) measures the rarity of a term across all documents in the corpus. It is calculated using the formula:

$$IDF(t) = \log \left(\frac{\text{Total number of documents}}{\text{Number of documents containing term } t} \right)$$

For example, let's calculate IDF for the term "cat":

$$IDF("cat") = \log \left(\frac{3}{2} \right) = 0.176$$

Similarly, we calculate IDF for all terms in the corpus.

Step 4: Calculate TF-IDF:

Finally, we calculate the TF-IDF score for each term in each document by multiplying its TF by its IDF.

For example, let's calculate TF-IDF for the term "cat" in Document 1:

$$TF\text{-}IDF("cat", \text{Document 1}) = TF("cat", \text{Document 1}) \times IDF("cat") = 0.167 \times 0.176 = 0.0294$$

Similarly, we calculate TF-IDF for all terms in each document.

The resulting TF-IDF scores represent the importance of each term in each document relative to the entire document collection. Terms with higher TF-IDF scores are considered more important and are often used as features in tasks such as document classification, clustering, and information retrieval.

b) With suitable example ,explain the steps involved in k-means algorithm

Ans:- The k-means algorithm is a popular unsupervised machine learning technique used for clustering data into k distinct groups or clusters based on similarity. It aims to partition the data into clusters in such a way that the within-cluster variation (inertia) is minimized. Here are the steps involved in the k-means algorithm:

****Step 1: Initialization****

1. ****Choose the Number of Clusters (k)****: Decide on the number of clusters you want to partition the data into. This is a crucial step and often requires domain knowledge or prior analysis of the data.

2. ****Initialize Cluster Centers****: Randomly select k data points from the dataset as the initial cluster centers or centroids. These centroids will serve as the starting points for the clusters.

****Step 2: Assign Data Points to Nearest Cluster****

3. ****Assign Data Points to Nearest Cluster****: For each data point in the dataset, calculate the distance (e.g., Euclidean distance) to each cluster center. Assign the data point to the cluster with the nearest centroid.

****Step 3: Update Cluster Centers****

4. ****Update Cluster Centers****: After assigning data points to clusters, calculate the new cluster centers by taking the mean of all data points assigned to each cluster. These new centroids represent the updated positions of the clusters.

****Step 4: Repeat Steps 2 and 3 Until Convergence****

5. ****Repeat Steps 2 and 3****: Iteratively repeat steps 2 and 3 until one of the convergence criteria is met. Common convergence criteria include:

- The centroids do not change significantly between iterations.
- The assignments of data points to clusters do not change between iterations.
- The maximum number of iterations is reached.

****Step 5: Finalize Clustering****

6. ****Finalize Clustering****: Once the algorithm converges, the final cluster assignments are obtained. Each data point belongs to one of the k clusters based on its proximity to the cluster centroids.

****Example: Clustering Iris Dataset with k-means****

Consider the famous Iris dataset, which contains measurements of sepal length, sepal width, petal length, and petal width for three species of iris flowers. Let's illustrate the steps of the k-means algorithm with this dataset:

1. ****Initialization****: Suppose we want to cluster the Iris dataset into 3 clusters ($k=3$). We randomly initialize three cluster centers.

2. ****Assign Data Points to Nearest Cluster****: For each data point, calculate the distance to each cluster center and assign it to the nearest cluster.

3. ****Update Cluster Centers****: Recalculate the centroids of the clusters based on the data points assigned to each cluster.
4. ****Repeat Steps 2 and 3****: Iterate steps 2 and 3 until convergence, i.e., until the cluster assignments and centroids stabilize or the maximum number of iterations is reached.
5. ****Finalize Clustering****: Once the algorithm converges, the final cluster assignments are obtained, and each data point is assigned to one of the three clusters.

Q6) a) Define following terms with respect to confusion matrix : [8]

i) Accuracy

ii) Precision

ii) Recall

iv) AUC-ROC

Ans:-

i) **Accuracy:**

Accuracy is a measure of the overall correctness of a classification model. It is calculated as the ratio of correctly classified instances (both true positives and true negatives) to the total number of instances in the dataset. In terms of the confusion matrix, accuracy can be calculated using the formula:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

- TP (True Positive): The number of instances that are truly positive and are classified as positive.
- TN (True Negative): The number of instances that are truly negative and are classified as negative.
- FP (False Positive): The number of instances that are actually negative but are classified as positive (Type I error).
- FN (False Negative): The number of instances that are actually positive but are classified as negative (Type II error).

ii) **Precision:**

Precision measures the proportion of correctly classified positive instances (true positives) out of all instances classified as positive (both true positives and false positives). It assesses the

model's ability to avoid falsely classifying negative instances as positive. In terms of the confusion matrix, precision can be calculated using the formula:

$$\text{Precision} = \frac{TP}{TP+FP}$$

iii) **Recall:**

Recall, also known as sensitivity or true positive rate (TPR), measures the proportion of correctly classified positive instances (true positives) out of all actual positive instances in the dataset. It assesses the model's ability to capture all positive instances without missing any. In terms of the confusion matrix, recall can be calculated using the formula:

$$\text{Recall} = \frac{TP}{TP+FN}$$

iv) **AUC-ROC (Area Under the Receiver Operating Characteristic Curve):**

AUC-ROC is a performance metric used for binary classification models that evaluates the model's ability to discriminate between positive and negative classes across different threshold values. The ROC curve is a graphical plot of the true positive rate (recall) against the false positive rate (FPR) at various threshold settings. AUC-ROC represents the area under the ROC curve, with values ranging from 0 to 1. Higher AUC-ROC values indicate better model performance, with 0.5 indicating random classification and 1 indicating perfect classification.

b) Explain k-fold Cross Validation & Random Subsampling

Ans:-

****1. K-fold Cross-Validation:****

K-fold cross-validation is a resampling technique used to evaluate the performance of a machine learning model. It involves partitioning the dataset into k equal-sized subsets, or folds. The model is trained k times, each time using k-1 folds as the training set and one fold as the validation set. The performance of the model is then averaged across the k iterations to obtain a robust estimate of its performance.

Here's how k-fold cross-validation works:

- Divide the dataset into k equal-sized folds.
- For each iteration:
 - Use k-1 folds as the training set.

- Use the remaining fold as the validation set.
- Train the model on the training set and evaluate its performance on the validation set using a chosen evaluation metric (e.g., accuracy, precision, recall).
- Average the performance metrics across all k iterations to obtain the final performance estimate.

K-fold cross-validation provides a more reliable estimate of the model's performance compared to a single train-test split, as it uses multiple validation sets and ensures that each data point is used for validation exactly once.

****2. Random Subsampling:****

Random subsampling, also known as simple random sampling, is a technique used to create multiple train-test splits from a dataset for model evaluation. It involves randomly partitioning the dataset into a training set and a test set multiple times and evaluating the model's performance on each split.

Here's how random subsampling works:

- Randomly partition the dataset into a training set and a test set, typically using a specified ratio (e.g., 70% training, 30% test).
- Train the model on the training set and evaluate its performance on the test set using a chosen evaluation metric.
- Repeat the process multiple times, each time randomly sampling from the dataset to create a new train-test split.
- Average the performance metrics across all iterations to obtain the final performance estimate.

Random subsampling is a simpler alternative to k-fold cross-validation and is often used when computational resources are limited or when the dataset is large. However, it may result in higher variance in the performance estimates compared to k-fold cross-validation, especially with small datasets or when the train-test splits are not representative of the overall dataset.

Q7) a) With a suitable example, draw a Histogram, boxplot and explain its usages.

Ans:-

****Example: Exam Scores****

Suppose we have a dataset containing exam scores of students in a class. We want to visualize the distribution of exam scores and identify any outliers in the dataset.

1. **Histogram:**

A histogram is a graphical representation of the distribution of numerical data. It divides the data into bins or intervals and displays the frequency or count of observations falling into each bin using bars of varying heights.

```
```python
import matplotlib.pyplot as plt

Example exam scores
exam_scores = [65, 72, 78, 80, 82, 85, 88, 90, 92, 95, 98, 100, 102, 105]

Plotting histogram
plt.hist(exam_scores, bins=5, color='skyblue', edgecolor='black')
plt.title('Histogram of Exam Scores')
plt.xlabel('Exam Scores')
plt.ylabel('Frequency')
plt.grid(True)
plt.show()
```
```

In the histogram, each bar represents a range of exam scores (bin), and the height of the bar indicates the frequency of scores falling into that range. Histograms provide insights into the distribution, central tendency, and spread of the data.

2. **Boxplot:**

A boxplot (box-and-whisker plot) is a graphical representation of the distribution of numerical data through quartiles. It provides a visual summary of the central tendency, dispersion, and skewness of the dataset, as well as the presence of outliers.

```
```python
Plotting boxplot
plt.boxplot(exam_scores, vert=False, patch_artist=True, boxprops=dict(facecolor='lightgreen'))
plt.title('Boxplot of Exam Scores')
plt.xlabel('Exam Scores')
plt.grid(True)
plt.show()
```
```

In the boxplot, the central box represents the interquartile range (IQR), with the median (50th percentile) indicated by the line inside the box. The whiskers extend to the minimum and

maximum values within 1.5 times the IQR from the lower and upper quartiles, respectively. Outliers beyond the whiskers are plotted individually as points.

****Usages:****

- ****Histogram Usage:****

- Identifying the distribution of a dataset: Histograms help visualize the shape and characteristics of the distribution, such as symmetry, skewness, or multimodality.
- Analyzing central tendency and spread: Histograms provide insights into the mean, median, and variability of the data.
- Detecting outliers: Histograms can reveal the presence of outliers or unusual patterns in the data.

- ****Boxplot Usage:****

- Summarizing distribution and variability: Boxplots provide a concise summary of the central tendency, spread, and skewness of the dataset.
- Identifying outliers: Boxplots visually highlight outliers as individual points beyond the whiskers, making them easy to identify.
- Comparing multiple groups: Boxplots allow for easy comparison of the distribution of multiple groups or categories within the dataset.

b) Describe the data visualization tool Tableau. List of data visualization tools.

Ans:- Tableau is a powerful and widely used data visualization tool that allows users to create interactive and visually appealing dashboards and reports from various data sources. It provides a user-friendly interface with drag-and-drop functionality, making it accessible to users with varying levels of technical expertise. Here's an overview of Tableau's key features:

1. **Data Connection:** Tableau supports seamless connectivity to a wide range of data sources, including databases, spreadsheets, cloud services, and web data connectors. Users can easily connect to their data sources and extract, transform, and analyze data without writing complex queries.

2. **Visualization:** Tableau offers a rich set of visualization options, including bar charts, line charts, scatter plots, maps, histograms, and more. Users can customize the appearance of visualizations by adjusting colors, fonts, sizes, and labels to create visually appealing and informative dashboards.

3. **Drag-and-Drop Interface:** Tableau's intuitive drag-and-drop interface allows users to quickly create and customize visualizations without the need for coding or scripting. Users can

simply drag fields from their data source onto the canvas to create visualizations and easily rearrange or resize them as needed.

4. ****Interactivity****: Tableau enables users to create interactive dashboards that allow viewers to explore and interact with the data dynamically. Users can add filters, parameters, and actions to dashboards to enable drill-down, filtering, and highlighting of data based on user interactions.

5. ****Publishing and Sharing****: Tableau provides options for publishing and sharing dashboards and reports with others. Users can publish their visualizations to Tableau Server or Tableau Online, where they can be securely accessed and shared with colleagues, clients, or stakeholders. Tableau also supports embedding visualizations in websites or sharing them via email.

6. ****Advanced Analytics****: Tableau offers built-in analytics capabilities, including trend analysis, forecasting, clustering, and statistical functions, allowing users to perform advanced data analysis directly within the tool.

7. ****Scalability and Performance****: Tableau is designed to handle large datasets and support scalable deployments for enterprise-level use. It offers features such as data caching, parallel processing, and in-memory data engine to optimize performance and ensure responsive visualization rendering.

Other Data Visualization Tools:

1. ****Microsoft Power BI****: A business analytics tool that enables users to create interactive reports and dashboards from various data sources.

2. ****Google Data Studio****: A free data visualization tool that allows users to create interactive dashboards and reports using data from Google products and other sources.

3. ****QlikView/Qlik Sense****: Business intelligence and data visualization platforms that offer powerful analytics and visualization capabilities for exploring and analyzing data.

4. ****D3.js (Data-Driven Documents)****: A JavaScript library for creating dynamic and interactive data visualizations in web browsers using HTML, SVG, and CSS.

5. ****Plotly****: A Python-based library for creating interactive and publication-quality visualizations, including charts, graphs, and dashboards.

6. ****Matplotlib****: A Python library for creating static, animated, and interactive visualizations in Python scripts, notebooks, and applications.

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Q8) a) What is Data Visualization? Describe the challenges of data visualization.

Ans:- Data visualization is the graphical representation of data to provide insights, facilitate understanding, and communicate patterns, trends, and relationships in the data. It involves creating visual representations such as charts, graphs, maps, and dashboards to present complex datasets in an intuitive and accessible manner.

****Challenges of Data Visualization:****

1. ****Data Complexity****: One of the primary challenges of data visualization is dealing with large and complex datasets. Visualizing high-dimensional data or datasets with millions of records can be challenging and may require advanced techniques for data aggregation, summarization, and visualization.
2. ****Interpretation****: Interpreting visualizations accurately can be challenging, especially when dealing with complex or unfamiliar data. Misinterpretation of visualizations can lead to erroneous conclusions or decisions, highlighting the importance of clear labeling, annotations, and context.
3. ****Choosing the Right Visualization****: Selecting the most appropriate visualization technique for a given dataset and analysis task is crucial for effective communication and interpretation of data. Choosing the wrong visualization type or misrepresenting data can lead to confusion and misinterpretation.
4. ****Biases and Misleading Visualizations****: Data visualizations can be subject to biases, inaccuracies, and misleading representations if not created and interpreted carefully. Biases may arise from data sampling, aggregation, or visualization design choices, leading to skewed or incorrect conclusions.
5. ****Technical Challenges****: Building interactive and dynamic visualizations that are responsive, scalable, and visually appealing can be technically challenging. Addressing performance issues, compatibility with different devices and browsers, and ensuring accessibility for users with disabilities are important considerations.
6. ****Data Quality and Integrity****: Visualizations are only as good as the underlying data they represent. Poor data quality, missing values, outliers, and inconsistencies can undermine the credibility and reliability of visualizations, highlighting the importance of data preprocessing and validation.
7. ****Communication and Storytelling****: Effectively communicating insights and findings through visualizations requires storytelling skills, narrative structure, and context. Presenting data in a meaningful and engaging way that resonates with the target audience can be challenging.
8. **Choosing the Right Visualization**: One of the primary challenges in data visualization is selecting the most appropriate visualization type for the data and the intended message. With a myriad of chart types available, such as bar charts, line charts, pie charts, scatter plots, and

more, choosing the right visualization that effectively communicates the insights can be challenging.

9. Data Quality and Integrity: Data visualization is only as good as the quality and integrity of the underlying data. Challenges such as missing values, outliers, errors, inconsistencies, and biases in the data can affect the accuracy and reliability of visualizations, leading to incorrect interpretations and decisions.

10. Overplotting and Clutter: Overplotting occurs when multiple data points overlap in a visualization, making it difficult to discern individual data points or patterns. Similarly, visual clutter resulting from too many data points, labels, or annotations can obscure meaningful insights and overwhelm the viewer.

11. Scale and Complexity: Large datasets with high dimensionality and complexity pose challenges for visualization. Visualizing such datasets may require specialized techniques, interactive tools, or advanced visualization methods to effectively explore and understand the data.

b) Explain architecture of Apache-Pig

Ans:- Apache Pig is a high-level platform and scripting language built on top of Apache Hadoop for processing and analyzing large datasets. It provides a simple and concise way to express data transformations and operations, allowing users to write complex data processing workflows without the need for low-level MapReduce programming. The architecture of Apache Pig comprises several components that work together to execute data processing tasks efficiently. Here's an overview of the architecture of Apache Pig:

1. **Pig Latin Script:**

- Apache Pig uses a scripting language called Pig Latin to express data processing operations. Pig Latin is a dataflow language that allows users to specify data transformations and operations in a concise and readable manner.

- Pig Latin scripts consist of a series of statements, or commands, that define the data processing workflow. These statements include loading data from sources, transforming data using operators and functions, and storing results to output locations.

2. **Pig Latin Compiler:**

- The Pig Latin compiler is responsible for parsing, analyzing, optimizing, and translating Pig Latin scripts into executable MapReduce jobs.

- When a Pig Latin script is submitted for execution, the compiler parses the script, builds an abstract syntax tree (AST), performs semantic analysis, and generates an optimized execution plan.

- The compiler optimizes the execution plan by applying various optimization techniques, such as predicate pushdown, filter pushdown, and join optimization, to improve performance and efficiency.

3. **Execution Engine:**

- The execution engine is responsible for executing the optimized execution plan generated by the Pig Latin compiler.

- Apache Pig supports multiple execution engines, including the MapReduce execution engine and the Tez execution engine. The choice of execution engine depends on the underlying processing framework and environment.

- In the MapReduce execution engine, the optimized execution plan is translated into a series of MapReduce jobs, which are then submitted to the Hadoop cluster for execution.

- In the Tez execution engine, the optimized execution plan is translated into a directed acyclic graph (DAG) of tasks, which are executed on the Tez framework for improved performance and resource utilization.

4. **Hadoop Distributed File System (HDFS):**

- Apache Pig interacts with the Hadoop Distributed File System (HDFS) to read input data from HDFS, store intermediate data, and write output data to HDFS.

- Input data is typically stored in HDFS as files or directories, which are accessed by Pig using Hadoop InputFormats.

- Intermediate data generated during the execution of Pig Latin scripts is stored in HDFS as temporary files or directories, which are managed by the Pig runtime.

5. **User-Defined Functions (UDFs):**

- Apache Pig supports user-defined functions (UDFs) written in Java, Python, or other programming languages, allowing users to extend the functionality of Pig by defining custom data processing functions.

- UDFs can be used to perform custom data transformations, calculations, aggregations, and other operations on input data.